4	**(a)**		$4! = 24$ and $4^4 = 256$ or $4! = 4 \times 3 \times 2 \times 1$ and $4^4 = 4 \times 4 \times 4 \times 4$ So $4! < 4^4$	**M1** **E1** **[2]**	**2.1** **2.2a**	Evaluating both expressions or clearly comparing the factors of each Clear conclusion seen	
4	**(b)**		Using counterexample $n = 1$ $1! = 1^1 = 1$ So the statement is false / Nina is incorrect	**M1** **E1** **[2]**	**2.1** **2.2a**	Attempt to find a counterexample Clear argument about the statement from $n = 1$	